When to Repair? Emulator-Based Maintenance Optimization of Reinforced Concrete Under Carbonation

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References

About Me

- Civil Engineer
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- Research funded by the Brazilian National Council for Scientific and Technological Development (CNPq)
- Host institution: University of Minnesota (USA)

The Maintenance Problem

Owners of aging reinforced concrete structures must decide **when to inspect** and **when to repair**, long before visible damage appears.

Repair too early

We pay for repairs the structure did not need yet.

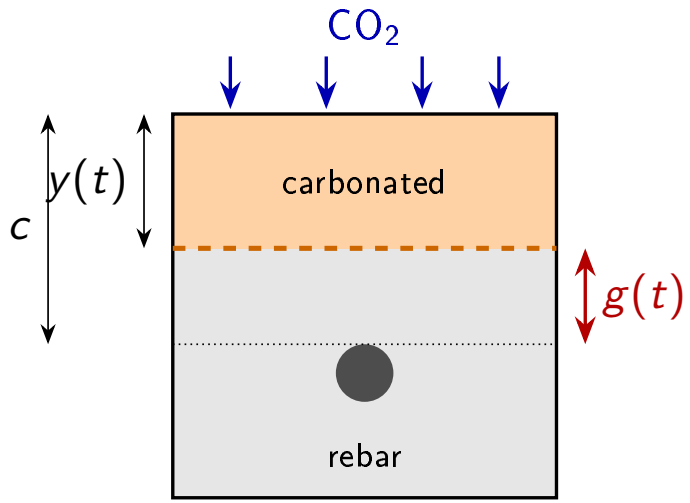
Repair too late

The steel is already corroding: the repair is more expensive and the risk is higher.

Research question

How often should we inspect, and at what condition should we repair, to minimize the life-cycle cost of a concrete element exposed to carbonation?

Carbonation and the Safety Margin



CO_2 penetrates the concrete and lowers its pH. When the carbonation front reaches the rebar, the steel loses its protection (*depassivation*).

Safety margin = distance to the rebar

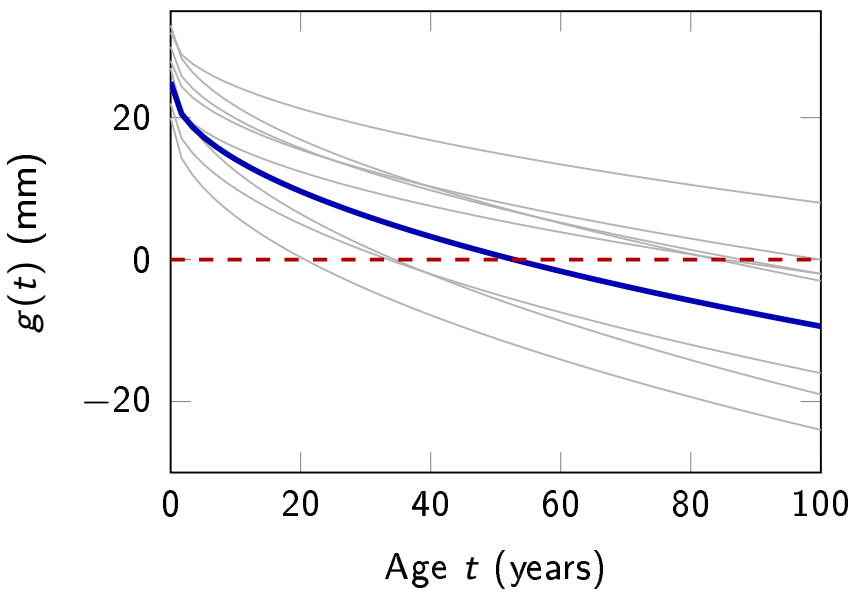
$$g(t) = c - y(t)$$

c: cover $\quad y(t)$: carbonation depth

Depassivation: $\quad g(t) \leq 0$

g can be measured in the field (mm).

The Margin Is Uncertain



Illustrative trajectories.

Concrete strength, humidity, cover and CO_2 are uncertain, so each element follows a different trajectory.

Probability of depassivation

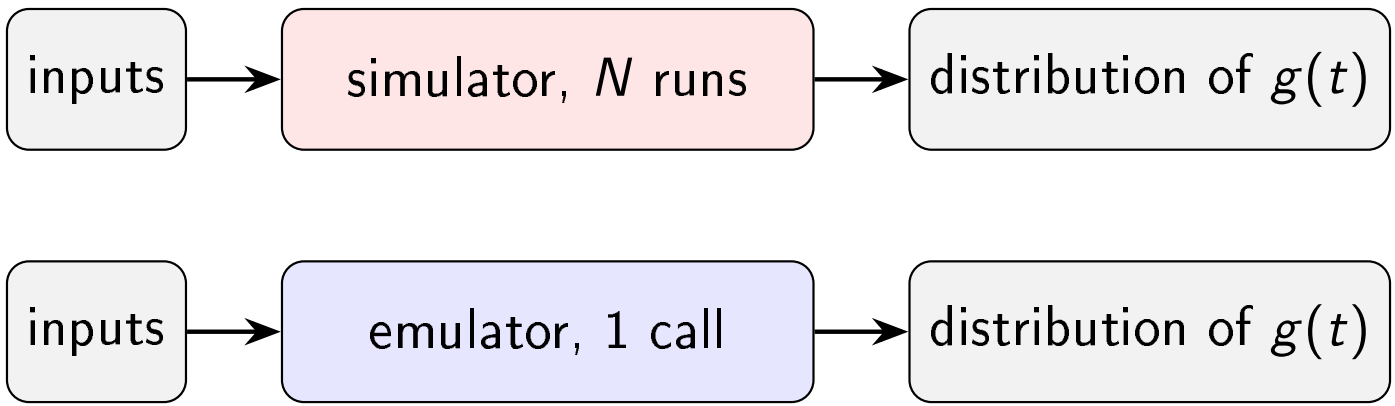
$$P_f(t) = P\left[g(t) \leq 0\right]$$

Example (25 MPa, 25 mm cover): without maintenance, $P_f(100) \approx 0.88$.

What Is an Emulator?

Emulator

A fast statistical model trained on runs of a simulator. It replaces the simulator whenever we need to call it many times [Sacks et al., 1989, Kennedy and O'Hagan, 2001].



Our emulator (neural network + Generalized Lambda Distribution) returns the **whole distribution** of $g(t)$ for a given design, at every age.

Why Use an Emulator Here?

Evaluating *one* maintenance plan means simulating thousands of elements over 100 years, with repairs. The optimizer tests *many* plans.

Without emulator

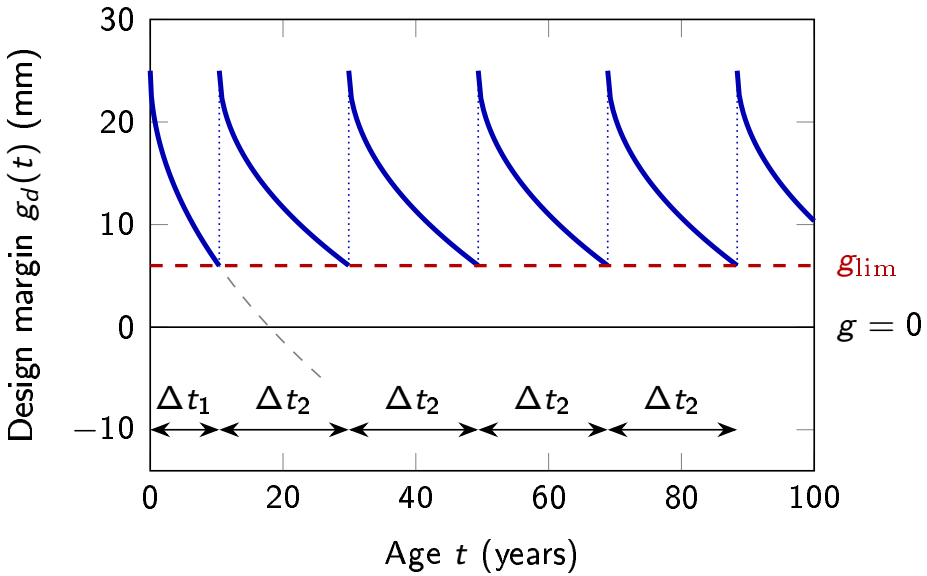
The physical model is called for every policy $\times$ every element $\times$ every repair cycle.

With emulator

The distribution of $g(t)$ is computed **once** per material (original concrete and repair mortar). All policies reuse it.

The same workflow applies to expensive models (finite elements, PDE solvers), where direct Monte Carlo is not feasible.

Maintenance Plan at the Design Stage



Illustrative: $c = 25$ mm, $g_{\lim} = 6$ mm. Gray dashed: no maintenance.

Design stage: nothing to inspect yet, but we **know the model**.

Rule

Repair when the design margin $g_d(t)$ reaches $g_{\lim}$.

Design margin

$g_d(t)$: 5% quantile of the emulator cloud (characteristic value).

Δt_1: original concrete.

Δt_2: repair mortar (different carbonation rate).

Decision Variable

g_{lim}: repair threshold (mm)

Distance from the carbonation front to the rebar that triggers a repair. It defines the whole maintenance schedule $t_1, t_2, \ldots$

Large g_{lim}

- Repair early
- Many repairs
- Few failures

Small g_{lim}

- Repair late
- Few repairs
- Many failures

Why a threshold and not a fixed interval?

The original concrete and the repair mortar carbonate at different rates. A threshold in mm gives a different interval for each cycle, and it is the language engineers use in the field.

Objective Function

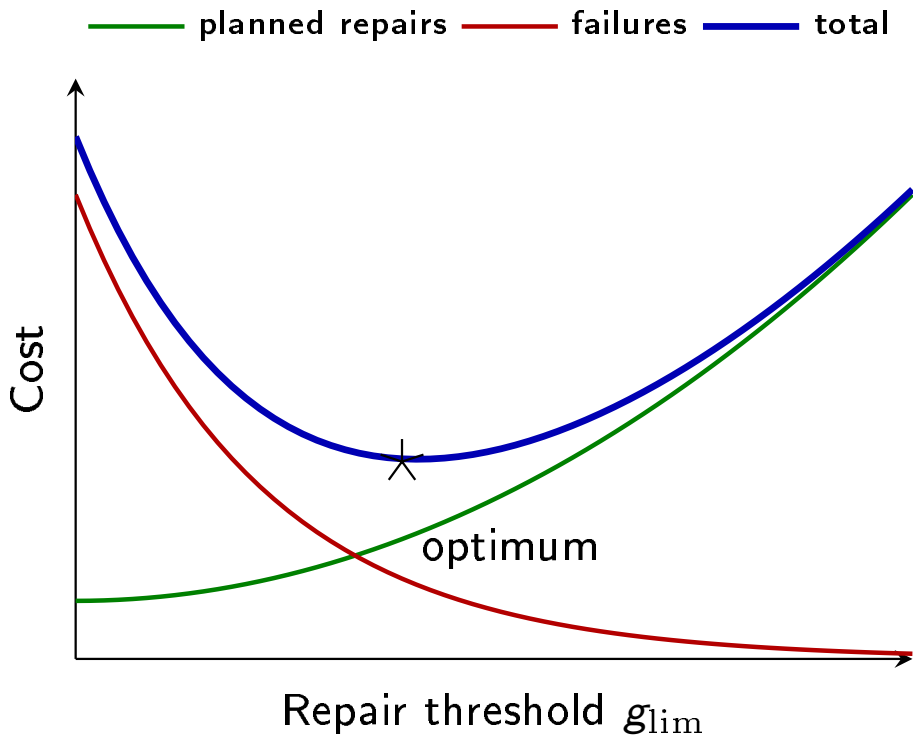
Expected life-cycle cost over $T = 100$ years

$$C_T(g_{\lim}) = \underbrace{c_{\text{rep}}\, N_{\text{rep}}(g_{\lim})}_{\text{planned repairs}} + \underbrace{c_{\text{fail}}\, \mathbb{E}\left[N_{\text{fail}}(g_{\lim})\right]}_{\text{failures in service}}$$

- N_{rep}: number of planned repairs (known once $g_{\lim}$ is fixed).
- N_{fail}: times the *real* margin reaches $g(t) = 0$ before the planned repair (random, from the emulator cloud).
- A failure in service triggers a corrective repair and the element restarts.
- Costs relative to C_0, the construction cost of the element.

Cost	Meaning	Initial value
c_{rep}	planned (preventive) repair	$0.10\, C_0$
c_{fail}	corrective repair + indirect costs	$0.50\, C_0$

Expected Trade-Off



Qualitative sketch (no results yet).

- **Small** $g_{\lim}$: few planned repairs, many failures.
- **Large** $g_{\lim}$: few failures, many planned repairs.
- We expect an **interior optimum**.
- The optimum depends on the ratio $c_{\text{fail}}/c_{\text{rep}}$.

Optimization Problem

Formulation

$$\min_{g_{\text{lim}}} \quad c_{\text{rep}}\, N_{\text{rep}}(g_{\text{lim}}) + c_{\text{fail}}\, \mathbb{E}\left[N_{\text{fail}}(g_{\text{lim}})\right]$$

$$\text{subject to} \quad 0 \leq g_{\text{lim}} \leq c_{\text{nom}}$$

How each evaluation works

1. Build the schedule $t_1, t_2, \ldots$ from the design margin $g_d(t)$ and g_{lim}.
2. Simulate many virtual elements from the emulator cloud over 100 years with this schedule.
3. Count the failures in service and average them: $\mathbb{E}[N_{\text{fail}}]$.

One variable: a simple scan over g_{lim} is enough to find the optimum.

Planned Parametric Study

Main study: how much does a failure cost?

	low	base	high		extreme
c_{fail}	$0.3\,C_0$	$0.5\,C_0$	$1.0\,C_0$	$2.0\,C_0$	$20\,C_0$
$c_{\text{fail}}/c_{\text{rep}}$	3	5	10	20	200

$c_{\text{rep}} = 0.10\,C_0$ fixed.

We will show

- Total cost $C_T(g_{\text{lim}})$, one curve per c_{fail}
- Optimal g^*_{lim} vs. $c_{\text{fail}}/c_{\text{rep}}$
- At the optimum: N_{rep}, $\mathbb{E}[N_{\text{fail}}]$ and the repair schedule

Secondary studies

- Cover c: 20, 25, 30, 40 mm
- Concrete strength f_c: 25, 35, 50 MPa
- Humidity RH: 55%, 75%
- Design quantile of g_d: 5%, 50%

Expected: the higher the failure cost, the larger g^*_{lim} and the more planned repairs.

Open Questions and Next Steps

Open questions

1. **Failure cost:** which range of $c_{\text{fail}}/c_{\text{rep}}$ is realistic for a serviceability failure?
2. **Design margin:** which quantile of the cloud defines $g_d(t)$ (5%, median)?
3. **Discounting:** should future costs be discounted to present value?
4. **End of horizon:** how to value the condition of the element at 100 years?

Next step: from design to operation

Once the structure is built, inspections measure the real margin and update the plan. The design-stage plan becomes the starting point; inspections tell us when to deviate from it.

References I



Kennedy, M. C. and O'Hagan, A. (2001).

Bayesian calibration of computer models.

Journal of the Royal Statistical Society: Series B (Statistical Methodology), 63(3):425–464.



Sacks, J., Welch, W. J., Mitchell, T. J., and Wynn, H. P. (1989).

Design and analysis of computer experiments.

Statistical Science, 4(4):409–423.

Thank you

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